TECHNICAL SPECIFICATIONS

SECTION 26 33 53

STATIC UNINTERRUPTIBLE POWER SUPPLY (UPS)

Project: Meridian Data Centre Campus, Phase 1
Navi Mumbai, India
Revision: 2024 (December 12, 2024)

PART 1 - GENERAL

1.1 Summary

This section specifies the technical requirements for static, double-conversion online uninterruptible power supply (UPS) modules to serve the critical IT load.

1.2 References

- IEC 62040-3 - Uninterruptible power systems (UPS) - Performance
- UL 1778 - Standard for Uninterruptible Power Systems
- IS 16242 - Static Uninterruptible Power Systems

1.3 Submittal Requirements

A. Contractor shall submit the following for Engineer's approval prior to equipment procurement:

1. Product Data: Technical datasheets detailing all electrical and physical parameters, performance curves, and compliance.
2. Shop Drawings: Dimensional layouts showing clearance requirements, cabling paths, and weights.
3. Compliance Matrix: Clause-by-clause confirmation of conformance. Any deviation must be explicitly noted.
4. Factory Test Certificates and Type Test Certifications.

1.4 Quality Assurance

A. Equipment shall be manufactured by a firm specializing in the equipment class with a minimum of twenty (25) years of continuous commercial manufacturing history.

B. As specified in Section 26 05 00 Part 1.4.3.B, all medium voltage switchgear and high-power electrical components must be supplied with certified test certificates from an internationally recognized independent laboratory (ASTA, KEMA, or equivalent) confirming compliance with IEC standards.

PART 2 - PRODUCTS

2.1 Acceptable Manufacturers

A. Subject to compliance with technical requirements, approved manufacturers are limited to the following:

2.2 Performance and Design Requirements

efficiency: UPS module double-conversion efficiency must be >= 96.0% at 50%, 75%, and 100% load points under online VFI mode.

redundancy: UPS modules must be configured as N+1 parallel redundant per group.

battery_runtime: Batteries must support a minimum of 10.0 minutes of autonomous runtime at 100% rated module load.

input_thd: Input current Total Harmonic Distortion (THDi) must be <= 5.0% at full load.

Injected Contract Defects (Internal Quality Control Audit Targets):

- **DEF-001:** Design Basis says UPS N+1, but Section 26 33 53 Part 2.1.2.A states 'UPS system modules shall be configured in a minimum N+2 parallel redundant configuration'. (Location: 26 33 53 Part 2.1.2.A vs DBR Section 3.1)
- **DEF-002:** UPS spec Section 26 33 53 Part 2.3.1.E requires a 10-min battery runtime, but its own Part 3.4.2.C Field testing acceptance criterion says 'inverter shall maintain full 100% IT load for a minimum of 12 minutes on battery power'. (Location: 26 33 53 Part 2.3.1.E vs Part 3.4.2.C)

PART 3 - EXECUTION

3.1 Installation and Examination

A. Verify that structural concrete floor slab is fully cured. Verify structural load limit of 2,800 kg/m^2 is not exceeded in UPS or battery rooms.

B. Maintain a minimum clear work clearance of 1000 mm in front of all enclosures, and 1200 mm above, in accordance with local Indian Electricity Rules and IEC standards.

3.2 Field Quality Control and Site Commissioning

A. Startup services shall be performed by factory-authorized technicians.

B. Commissioning tests must be scheduled and carried out in accordance with the project's overall Cx Test Register. Every testing step must map back to a specific specification clause.

efficiency_verification: Conduct load bank testing to verify efficiency is >= 96.0% (and >= 96.5% after Addendum 3) at all load points.

battery_discharge: Verify battery autonomous runtime is >= 10 minutes at full load.